QUESTION 15: MALFORMED FIXED-WIDTH RECORD EVALUATION

AIM

To write a COBOL program that validates a transaction date in YYYYMMDD format, rejects invalid months and days, handles leap years correctly, and displays whether the date is accepted or rejected.

ALGORITHM

1. Start the program.
2. Read the date in YYYYMMDD format.
3. Check whether all characters are numeric; otherwise, display INVALID DATE FORMAT.
4. Extract the year, month, and day from the date.
5. Check that the month is between 1 and 12; otherwise, display INVALID MONTH.
6. Determine the maximum valid day for the selected month.
7. For February, allow 29 days in a leap year (divisible by 400, or by 4 but not by 100); otherwise, allow 28 days.
8. If the day is outside the valid range, display INVALID DAY.
9. If all checks pass, display ACCEPTED.
10. Stop the program.

PROGRAM

```
IDENTIFICATION DIVISION.
PROGRAM-ID. DATEVAL.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 WS-DATE PIC X(8).
01 WS-YEAR PIC 9(4).
01 WS-MON PIC 99.
01 WS-DAY PIC 99.
01 WS-MAX PIC 99 VALUE 0.
PROCEDURE DIVISION.
    DISPLAY "ENTER DATE (YYYYMMDD):"
    ACCEPT WS-DATE
    IF WS-DATE IS NUMERIC
        MOVE WS-DATE(1:4) TO WS-YEAR
        MOVE WS-DATE(5:2) TO WS-MON
        MOVE WS-DATE(7:2) TO WS-DAY
        IF WS-MON < 1 OR WS-MON > 12
            DISPLAY "INVALID MONTH"
        ELSE
            EVALUATE WS-MON
                WHEN 1
                WHEN 3
                WHEN 5
                WHEN 7
                WHEN 8
                WHEN 10
                WHEN 12
                    MOVE 31 TO WS-MAX
                WHEN 4
                WHEN 6
                WHEN 9
                WHEN 11
                    MOVE 30 TO WS-MAX
                WHEN 2
                    MOVE 28 TO WS-MAX
                    IF FUNCTION MOD(WS-YEAR, 400) = 0
                        MOVE 29 TO WS-MAX
                    ELSE
                        IF FUNCTION MOD(WS-YEAR, 4) = 0
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                        AND FUNCTION MOD(WS-YEAR, 100)
                        NOT = 0
                        MOVE 29 TO WS-MAX
                    END-IF
                END-IF
        END-EVALUATE
        IF WS-DAY < 1 OR WS-DAY > WS-MAX
            DISPLAY "INVALID DAY"
        ELSE
            DISPLAY "ACCEPTED"
        END-IF
    END-IF
ELSE
    DISPLAY "INVALID DATE FORMAT"
END-IF
STOP RUN.
```
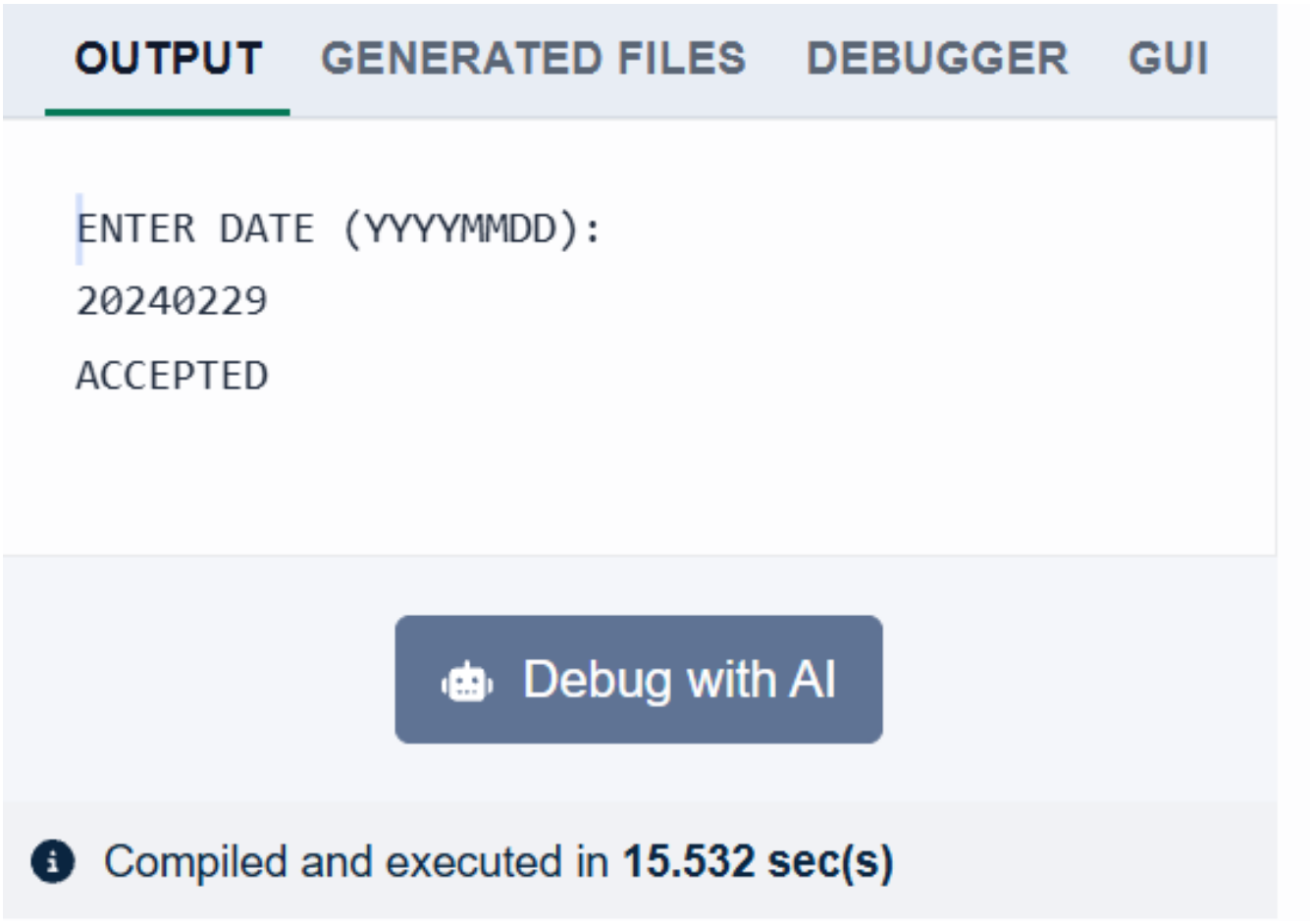

OUTPUT


OUTPUT
GENERATED FILES
DEBUGGER
GUI
ENTER DATE (YYYYMMDD):
20240229
ACCEPTED
Debug with AI
Compiled and executed in 15.532 sec(s)